

Low-Dimensional, Rank-One, and First Higher-Cycle Closures of the Pinching Refinement Conjecture

Automated mathematics run `fa_banach_001`, lane 3

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Abstract

Dinh asked whether the normalized trace average of all words with n letters A and m letters B dominates the same average after pinching B onto the commutant of A . We prove the conjecture (i) for all positive semidefinite 2×2 matrices and all $n, m \geq 0$, (ii) for rank-one A in every dimension and all $n, m \geq 0$, (iii) in every dimension when $n \leq 1$, and (iv) in every dimension at the first higher-cycle corner $(n, m) = (2, 3)$. A more general path argument covers every pair for which, in an eigenbasis of A , a diagonal unitary gauges B to an entrywise nonnegative matrix. The general higher-dimensional, cyclic-phase case remains open. The status is *candidate partial result, likely valid, pending human review*.

1 Source question and notation

The source is Trung Hoa Dinh, *On the Failure of the Upper Bound in the Refined BMV Conjecture and a Pinching Correction*, arXiv:2605.17782 (2026) [1]. Conjecture 5.1, on PDF page 7, is reproduced in Figure 1. The source proves the conjecture when $m = 2$, but leaves the general case open.

For positive semidefinite matrices $A, B \in M_d(\mathbb{C})$, let

$$\mathcal{A}_{n,m}(A, B) = \frac{1}{\binom{n+m}{n}} \sum_W \text{Tr } W(A, B), \quad (1)$$

where the sum runs over all words containing exactly n letters A and m letters B . If $A = \sum_\lambda \lambda P_\lambda$ is the spectral decomposition, set

$$E_A(B) = \sum_\lambda P_\lambda B P_\lambda. \quad (2)$$

Because A and $E_A(B)$ commute, the source conjecture is

$$\mathcal{A}_{n,m}(A, B) \geq \mathcal{A}_{n,m}(A, E_A(B)) = \text{Tr}(A^n E_A(B)^m) \quad (n, m \geq 0). \quad (3)$$

Zero powers are identities. Thus the claim is: *for all positive semidefinite A, B and all nonnegative integers n, m , inequality (3) holds.*

2 Proof intuition

In an eigenbasis of A , expand the trace of a word as a sum over closed index paths. An A -step cannot change the current spectral vertex, while a B -step may. Pinching deletes precisely the

creates spectral bridges, and the mixed words capture more of this bridge-gain than the clustered word does.

5 The corrected pinching conjecture

The previous sections suggest that the correct refinement should not be an upper bound by a clustered word. It should be a lower bound by the pinched commuting part.

Conjecture 5.1 (Pinching refinement). *Let $A, B \geq 0$. Then, for all $n, m \geq 0$,*

$$\mathcal{A}_{n,m}(A, B) \geq \mathcal{A}_{n,m}(A, E_A(B)). \quad (5.1)$$

Equivalently,

$$\mathcal{A}_{n,m}(A, B) \geq \text{Tr}(A^n E_A(B)^m). \quad (5.2)$$

Since A commutes with $E_A(B)$, the right-hand side is nonnegative. Therefore Conjecture 5.1 implies the coefficient positivity part of the BMV theorem:

$$\mathcal{A}_{n,m}(A, B) \geq 0.$$

The conceptual difference from the failed upper bound is important. The failed inequality compares the word average with one clustered word. The pinching conjecture compares the word average with the part already visible in the commutative algebra generated by A .

Figure 1: Conjecture 5.1 and its equivalent form in the source paper, cropped from PDF page 7.

B -edges that cross between distinct eigenspaces of A . Thus the desired difference is the sum of weights of closed paths that use at least one deleted edge.

This is not a proof in general: in dimension at least three, a closed B -cycle can carry a nontrivial complex phase and hence have negative real part. It does prove the claim when all B -entries can be made nonnegative by a diagonal phase gauge. Every 2×2 pair has this property, because a single off-diagonal phase can always be removed.

Two further arguments avoid path phases. If A has rank one, cyclicity rewrites the word average as an average over weak compositions of m ; rank-one collapse then reduces each term to scalar moments of B , which dominate the corresponding powers of the first moment. If $n \leq 1$, the same scalar moment inequality applied to compressions of B proves the claim in every dimension. At $(n, m) = (2, 3)$, cyclicity leaves just two trace expressions. One is another compression moment; the other becomes a squared-norm estimate after normalizing B to a correlation matrix. None of these mechanisms controls arbitrary higher-dimensional cyclic phases, so the original full conjecture remains open.

3 A compression moment lemma

Lemma 3.1 (compression moments). *Let $B \geq 0$, let P be an orthogonal projection, and let r be a nonnegative integer. Then*

$$\text{Tr}(PB^r P) \geq \text{Tr}((PBP|_{PC^d})^r). \quad (4)$$

Equivalently, with both sides viewed on $\mathbb{C}^d$, $\text{Tr}(PB^r P) \geq \text{Tr}((PBP)^r)$ for $r \geq 1$; for $r = 0$, the right-hand side is understood as the identity on PC^d .

Proof. Diagonalize the positive compression $C = PBP|_{P\mathbb{C}^d}$, say $Ce_j = \mu_j e_j$ for an orthonormal basis of $P\mathbb{C}^d$. If $B = \sum_k t_k Q_k$ is a spectral decomposition, then the nonnegative numbers $\langle e_j, Q_k e_j \rangle$ sum to one. Convexity of $t \mapsto t^r$ on $[0, \infty)$ gives

$$\langle e_j, B^r e_j \rangle = \sum_k t_k^r \langle e_j, Q_k e_j \rangle \geq \left(\sum_k t_k \langle e_j, Q_k e_j \rangle \right)^r = \mu_j^r.$$

(The cases $r = 0, 1$ are equalities.) Summing over j proves the lemma. $\square$

4 The path-positive class and the 2×2 theorem

Theorem 4.1 (phase-balanced pairs). *Suppose there is an orthonormal eigenbasis of A and a diagonal unitary D in that basis such that D^*BD is entrywise real and nonnegative. Then (3) holds for every $n, m \geq 0$.*

Proof. Conjugating both matrices by D changes neither side of (3); moreover D commutes with A , and pinching is covariant under this conjugation. We may assume that A is diagonal with nonnegative diagonal entries and every entry of B is nonnegative.

Fix $W = X_1 \cdots X_{n+m}$, with each X_s equal to A or B . Its trace has the closed-path expansion

$$\mathrm{Tr} W = \sum_{i_0, \dots, i_{n+m-1}} (X_1)_{i_0 i_1} (X_2)_{i_1 i_2} \cdots (X_{n+m})_{i_{n+m-1} i_0}. \quad (5)$$

Every summand is nonnegative. If $X_s = A$, its factor vanishes unless the path stays at the same vertex. Replacing B by $E_A(B)$ sets B_{ij} to zero exactly when the eigenvalues of A at i and j are different, and leaves all other entries unchanged. Hence pinching deletes a subset of nonnegative summands in (5). Thus $\mathrm{Tr} W(A, B) \geq \mathrm{Tr} W(A, E_A(B))$ word by word. Averaging proves the claim. $\square$

Corollary 4.2 (all 2×2 cases). *For all positive semidefinite $A, B \in M_2(\mathbb{C})$ and all $n, m \geq 0$, inequality (3) holds.*

Proof. Choose an eigenbasis of A . If A is scalar, then $E_A(B) = B$. Otherwise write

$$B = \begin{pmatrix} x & z \\ \bar{z} & y \end{pmatrix}, \quad x, y \geq 0.$$

A diagonal unitary makes z real and nonnegative, so Theorem 4.1 applies. Directly, a closed two-vertex path uses equally many $1 \rightarrow 2$ and $2 \rightarrow 1$ transitions, whose product is $|z|^{2q} \geq 0$. $\square$

Remark 4.3. *Repeated eigenvalues cause no problem: pinching retains all edges within each eigenspace and deletes only cross-eigenspace edges.*

5 Rank-one A in arbitrary dimension

Lemma 5.1 (cyclic composition formula). *For $n \geq 1$,*

$$\mathcal{A}_{n,m}(A, B) = \frac{1}{\binom{n+m-1}{m}} \sum_{r_1 + \cdots + r_n = m} \mathrm{Tr}(AB^{r_1} AB^{r_2} \cdots AB^{r_n}), \quad (6)$$

where the sum is over weak compositions $r_j \geq 0$.

Proof. Words beginning with A are in bijection with the weak compositions in (6). Mark one of the n occurrences of A in each unrestricted word and cyclically rotate that occurrence to the first position. If T is the sum over all words and S the sum over words beginning with A , this marked rotation is N -to-one, where $N = n + m$, even for periodic words (the marked position is retained). Cyclic invariance therefore gives $nT = NS$. Since $\frac{n}{N}\binom{N}{n} = \binom{N-1}{n-1} = \binom{n+m-1}{m}$, the unrestricted average equals the average over words beginning with A , proving the formula. $\square$

Theorem 5.2 (rank-one closure). *If $A \geq 0$ has rank one, then (3) holds in every finite dimension for all $n, m \geq 0$.*

Proof. The case $n = 0$ follows from Lemma 3.1, summed over the spectral projections of A , and is also in Theorem 6.1. Assume $n \geq 1$, and write $A = aP$, where $a > 0$ and $P = |v\rangle\langle v|$ for a unit vector v . Put $\mu = \langle v, Bv \rangle$. Rank-one collapse in (6) gives

$$\mathrm{Tr}(AB^{r_1} \cdots AB^{r_n}) = a^n \prod_{j=1}^n \langle v, B^{r_j} v \rangle. \quad (7)$$

For every nonnegative integer r , the scalar spectral-measure argument in Lemma 3.1 gives $\langle v, B^r v \rangle \geq \mu^r$. Therefore every summand in (6) is at least $a^n \mu^m$, so

$$\mathcal{A}_{n,m}(A, B) \geq a^n \mu^m. \quad (8)$$

The spectral projections of A are P and $I - P$, hence $E_A(B) = \mu P + (I - P)B(I - P)$, and consequently

$$\mathrm{Tr}(A^n E_A(B)^m) = a^n \mu^m.$$

Together with (8), this proves the claim. $\square$

Remark 5.3. Rank-one A also follows from Theorem 4.1: diagonalize the compression of B to $v^\perp$, then adjust eigenvector phases so that all entries coupling them to v are nonnegative. The composition proof gives an independent scalar-moment mechanism.

6 The range $n \leq 1$ in arbitrary dimension

Theorem 6.1 (zero or one A -letter). *For arbitrary positive semidefinite matrices A, B in every finite dimension, (3) holds for $n = 0$ and $n = 1$, for every $m \geq 0$.*

Proof. Let $A = \sum_\lambda \lambda P_\lambda$, and write $C_\lambda = P_\lambda B P_\lambda|_{P_\lambda \mathbb{C}^d}$. If $n = 0$, Lemma 3.1 gives

$$\mathcal{A}_{0,m}(A, B) = \mathrm{Tr}(B^m) = \sum_\lambda \mathrm{Tr}(P_\lambda B^m P_\lambda) \geq \sum_\lambda \mathrm{Tr}(C_\lambda^m) = \mathrm{Tr}(E_A(B)^m).$$

If $n = 1$, cyclic invariance makes every word trace equal to $\mathrm{Tr}(AB^m)$. Applying Lemma 3.1 blockwise and weighting by the nonnegative eigenvalues λ yields

$$\begin{aligned} \mathcal{A}_{1,m}(A, B) &= \mathrm{Tr}(AB^m) = \sum_\lambda \lambda \mathrm{Tr}(P_\lambda B^m P_\lambda) \\ &\geq \sum_\lambda \lambda \mathrm{Tr}(C_\lambda^m) = \mathrm{Tr}(A E_A(B)^m). \end{aligned}$$

$\square$

7 The first higher-cycle corner

Theorem 7.1 (the case $(n, m) = (2, 3)$). *For arbitrary positive semidefinite matrices A, B in every finite dimension,*

$$\mathcal{A}_{2,3}(A, B) \geq \text{Tr}(A^2 E_A(B)^3).$$

Proof. Choose an eigenbasis of A , and within each eigenspace choose a basis that diagonalizes the corresponding compression of B . Then $D = E_A(B)$ is diagonal in this basis. The cyclic-composition formula, now marking one of the three B -letters, gives

$$\mathcal{A}_{2,3}(A, B) = \frac{1}{6} \sum_{p+q+r=2} \text{Tr}(BA^p BA^q BA^r) = \frac{1}{2} \{ \text{Tr}(A^2 B^3) + \text{Tr}(ABAB^2) \}.$$

Lemma 3.1, applied on every spectral block of A and weighted by the corresponding eigenvalue squared, yields

$$\text{Tr}(A^2 B^3) \geq \text{Tr}(A^2 D^3). \quad (9)$$

It remains to prove the same lower bound for the second trace. Write $d_i = b_{ii}$. If $d_i = 0$, positivity of B forces its i -th row and column to vanish, so those indices may be discarded. On the remaining indices put

$$C = D^{-1/2} B D^{-1/2}, \quad X = AD.$$

Then $C \geq 0$, $c_{ii} = 1$, and $X = \text{diag}(x_i) \geq 0$ with $x_i = a_i d_i$. Direct substitution and cyclicity give

$$\begin{aligned} \text{Tr}(ABAB^2) &= \text{Tr}(XCXCDC) \\ &= \sum_k d_k (CXCXC)_{kk}. \end{aligned} \quad (10)$$

For each k ,

$$(CXCXC)_{kk} = \left\| C^{1/2} X C e_k \right\|^2.$$

Since $\|C^{1/2} e_k\|^2 = c_{kk} = 1$, Cauchy–Schwarz gives

$$\begin{aligned} \left\| C^{1/2} X C e_k \right\|^2 &\geq \left| \left\langle C^{1/2} e_k, C^{1/2} X C e_k \right\rangle \right|^2 \\ &= \left(\sum_j x_j |c_{jk}|^2 \right)^2 \geq x_k^2. \end{aligned}$$

Substitution in (10) proves

$$\text{Tr}(ABAB^2) \geq \sum_k d_k x_k^2 = \sum_k a_k^2 d_k^3 = \text{Tr}(A^2 D^3).$$

Combining this with (9) and the displayed average identity proves the theorem. $\square$

8 Verification and adversarial checks

The script `code/check_pinching_subcases.py` independently computes word averages as coefficients of $\text{Tr}(A + tB)^{n+m}$, constructs spectral pinching, and checks random complex instances of all

proved classes. It is a sanity check, not part of any proof. It covers random 2×2 pairs, arbitrary-dimensional $n = 0, 1$, random rank-one A , phase-balanced pairs, the all-dimensional $(n, m) = (2, 3)$ theorem, and the source’s known $m = 2$ range.

A bounded counterexample search preceded promotion: random real and complex PSD matrices in dimensions 3, 4, 5, ranks from one to full, total word degree up to 20, continuous optimization over Gram factors, and structured correlation-matrix families. No negative gap was found. This is evidence only. Convexity in B was tested and rejected as a proof route: the Hessian of the full coefficient can have negative directions, so a direct Jensen argument for the full word average is unavailable.

The formal proofs have these independent failure checks.

- Theorem 4.1 keeps, rather than deletes, edges inside repeated eigenspaces.
- In Corollary 4.2, Hermitian symmetry makes the two directed off-diagonal factors conjugates.
- The normalization in (6) is exactly the number $\binom{n+m-1}{m}$ of weak compositions.
- Lemma 3.1 uses scalar convexity in spectral measures, not the false claim that t^r is operator convex for every r .
- Theorem 7.1 removes zero diagonal entries before using $D^{-1/2}$, and its correlation estimate is a squared norm rather than a termwise assertion about signed three-cycles.

9 Scope, novelty, and review

This packet does *not* prove Conjecture 5.1 in full. It closes the first higher-cycle corner $(n, m) = (2, 3)$, but leaves open the remaining cases with $d \geq 3$, $n \geq 2$, $m \geq 3$ and genuinely unbalanced cycle phases. The source already proves $m = 2$ in all dimensions; that known case is not claimed as new.

A bounded novelty search on 11 August 2026 used the exact arXiv id, title, the phrases *pinching refinement* and *BMV pinching conjecture*, and exact text from Conjecture 5.1. It checked the local run indexes and arXiv/web results. It found the source and literature on the failed refined-BMV upper bound, but no later proof, counterexample, or statement of the 2×2 , rank-one- A , $n \leq 1$, phase-balanced, or $(n, m) = (2, 3)$ closures. This bounded search is not a guarantee of global novelty.

Recommended human review: *promote as a likely-valid partial result after checking the path-deletion identification in Theorem 4.1 the marked-word count in Lemma 5.1, and the correlation normalization in Theorem 7.1.* The genuinely open next target is control of signed higher-dimensional cycle phases.

References

- [1] T. H. Dinh, *On the Failure of the Upper Bound in the Refined BMV Conjecture and a Pinching Correction*, arXiv:2605.17782 (2026), <https://arxiv.org/abs/2605.17782>.